Stovetop Monitor User Manual

By: Jim Schrempp and Bob Glicksman; v1.1, 8/10/2026

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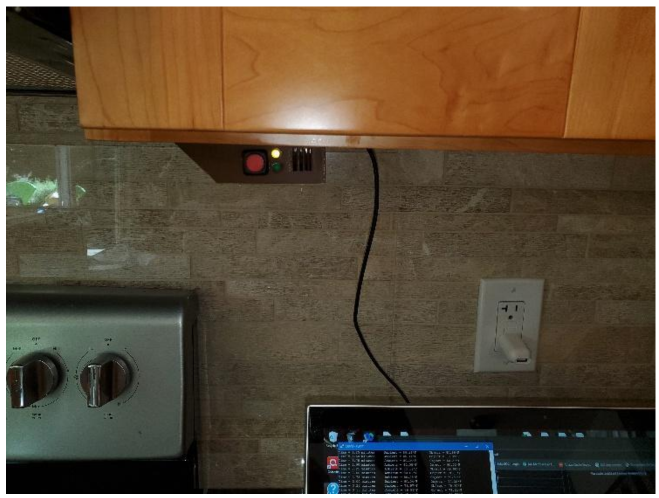
https://github.com/TeamPracticalProjects/Stovetop_Monitor/tree/main/Documents/Terms_of_Use_License_and_Disclaimer.pdf



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DOCUMENT OVERVIEW.

This document describes how to use the Stovetop Monitor. The Stovetop Monitor monitors the state of a stovetop and alarms if it senses that a burner is turned on for too long. The Stovetop Monitor has local controls and indicators as well as an App for remote monitoring and alarm reset.

STOVETOP MONITOR LOCAL CONTROLS AND INDICATORS.

The front of the Stovetop Monitor enclosure is shown in figure 1, below. The following devices are available to the user:

- A green LED.
- A yellow LED.
- A red backlit pushbutton switch.
- A grill to conduct the sound of an internal buzzer out the front of the enclosure.

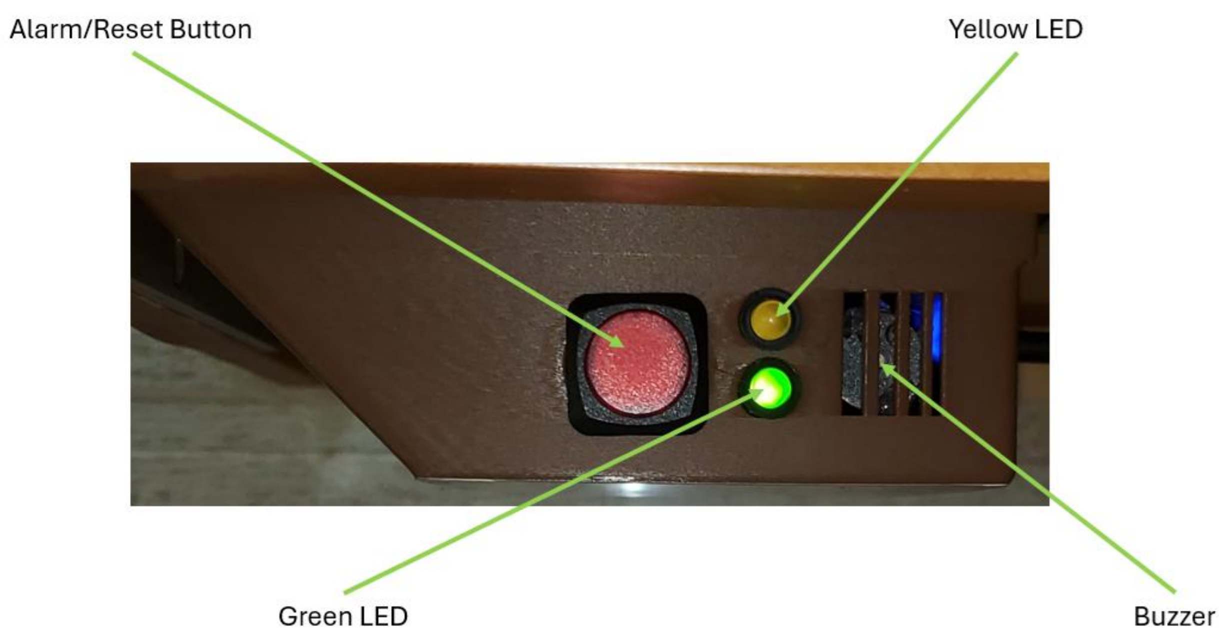

Figure 1. Stovetop Monitor Local Indicators and Control.

The following indications inform the user of the state of the stovetop being monitored:

1. Green LED on steadily: This indicates that the Stovetop Monitor is in the NORMal state. The NORMal state is when the IR temperature is near or below the ambient temperature, indicating that none of the burners are on.

2. Yellow LED on steadily: This indicates that the Stovetop Monitor is in the WARMing state. The WARMing state is when the IR temperature is higher than the ambient temperature by only a few degrees, indicating that one or more burners are on low for keeping food warm.

3. Yellow LED flashing: This indicates that the Stovetop Monitor is in the COOKing state. The COOKing state is when the IR temperature is higher than the ambient temperature by more than a few degrees, indicating that one or more burners are on medium / high for cooking food.

4. Red LED (backlight of the pushbutton) on steadily: This indicates that the Stovetop Monitor is in the BURNing state. The BURNing state is when the IR temperature is higher than normal for cooking, indicating that one or more burners are on medium /high and uncovered or else that one or more burners are on medium / high and the food has boiled down to where it is starting to burn. This is an *abnormal* condition and must be rectified within a few minutes or the alarm will be triggered.

5. Red LED (backlight of the pushbutton) flashing and Buzzer beeping: This indicates that the Stovetop Monitor has timed out in either the WARMing, COOKing, or BURNing state and an alarm condition exists. The alarm condition will persist until the user clears the alarm.

Each of the WARMing, COOKing, and BURNing states has a timeout. If the timeout expires while in one of these states, an alarm is generated and the alarm must be reset manually. The user can reset the alarm in one of two ways:

- By pressing the red pushbutton.
- By tapping the “Reset Alarm” button on the App (see below).

The timeouts and the thresholds for transitioning between states are contained in the file:

https://github.com/TeamPracticalProjects/Stovetop_Monitor/blob/main/Software/StovetopMonitor/src/ElectricStovetopConstants.h

The threshold values contain a temperature difference between the measured probe temperature and the ambient temperature. Each state has two threshold values, one for moving up to the next higher state (e.g. WARMing to COOKing) and a separate value for moving down to the next lower state (e.g. COOKing to WARMing). The probe is remote and the reported temperature values can vary slightly over time. The two individual values are needed to prevent the Stovetop Monitor from bouncing back and forth between states due to these small variations.

The values in this file were empirically determined by recording and analyzing various cooking and warming sessions on a particular electric glasstop stove. The type of stove, the location of the sensor, and the user's particular cooking styles will all influence the proper values to use in any particular installation. The Stovetop Monitor firmware writes timestamped values of ambient temperature and IR temperature to the serial port. Connecting the serial port to a computer running a serial monitor program allows the user to collect data about some cooking session which can be stored and subsequently analyzed to help the user develop installation specific timeouts and thresholds for their specific situation.

The following are the timeouts and thresholds that are supplied with this project release:

- **WARM_UP_TH**. If the IR temperature exceeds the ambient temperature by at least this amount, the Stovetop Monitor transitions from the NORMal state to the WARMing state. The value in the released firmware is 1.5 degrees F.
- **COOK_UP_TH**. If the IR temperature exceeds the ambient temperature by at least this amount, the Stovetop Monitor transitions from the WARMing state to the COOKing state. The value in the released firmware is 7.0 degrees F.
- **BURN_UP_TH**. If the IR temperature exceeds this value (regardless of ambient temperature), the Stovetop Monitor transitions from the COOKing state to the BURNing state. The value in the released firmware is 130.0 degrees F.

- **WARM_DN_TH**. If the IR temperature exceeds the ambient temperature by no more than this amount, the Stovetop Monitor transitions from the WARMing state to the NORMal state. The value in the released firmware is 1.0 degrees F.
- **COOK_DN_TH**. If the IR temperature exceeds the ambient temperature by no more than this amount, the Stovetop Monitor transitions from the COOKing state to the WARMing state. The value in the released firmware is 6.0 degrees F.
- **BURN_DN_TH**. If the IR temperature is less than this value (regardless of ambient temperature), the Stovetop Monitor transitions from the BURNing state to the COOKing state. The value in the released firmware is 125.0 degrees F.

- **WARM_ALARM_TIME**. If the Stovetop Monitor is in the WARMing state for longer than this time, it transitions to the ALARM state and an alarm is generated. The value in the released firmware is 65 minutes.
- **COOK_ALARM_TIME**. If the Stovetop Monitor is in the COOKing state for longer than this time, it transitions to the ALARM state and an alarm is generated. The value in the released firmware is 40 minutes.
- **BURN_ALARM_TIME**. If the Stovetop Monitor is in the BURNing state for longer than this time, it transitions to the ALARM state and an alarm is generated. The value in the released firmware is 5 minutes.

See the "Overview and Theory of Operation Manual" in this repository for further information about the operation of the Stovetop Monitor.

STOVETOP MONITOR APP.

A mobile App was developed as part of this project. This App allows the user to quickly determine the current state information about their Stovetop Monitor. Figure 2 shows the main screen of the App.

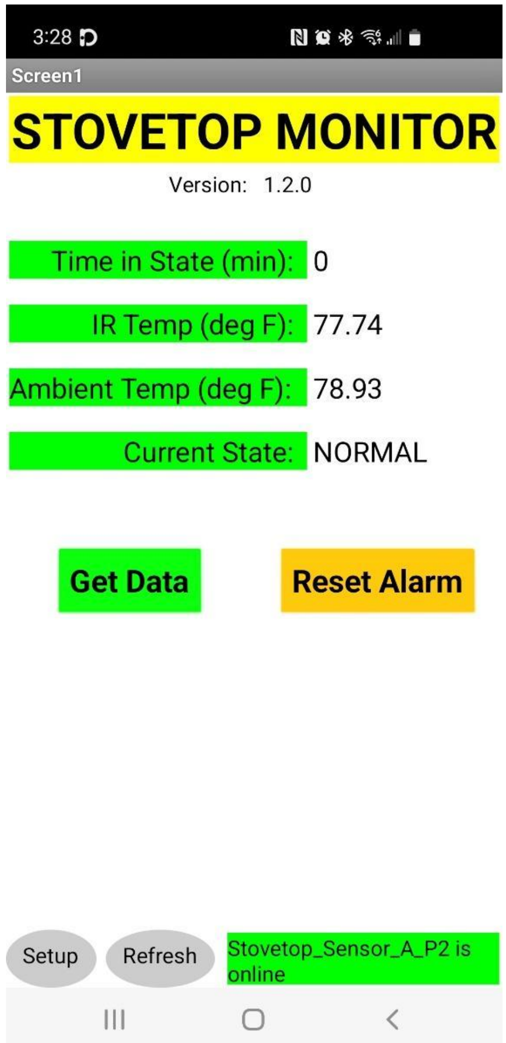

Figure 2. Stovetop Monitor Main App Screen.

When the App is opened and connected to a Stovetop Monitor device, all of the data shown on the main App screen (figure 2) is updated. The following data is displayed on this screen:

- **Version**. The version of the firmware that is running on the Stovetop Monitor.
- **Time in State (min)**. The number of minutes that the Stovetop Monitor is in the current state. In the NORMal state, this value is always zero.
- **IR Temp (deg F)**. The most recent IR temperature reading from the sensor.
- **Ambient Temp (deg F)**. The most recent ambient temperature reading from the sensor.

- **Current State**. The current state of the Stovetop Monitor (NORM, WARM, COOK, BURN, ALARM).

The user can tap the **Get Data** button at any time to update the data displayed to the very latest values.

In the event that the Stovetop Monitor is in the ALARM state, tapping the **Reset Alarm** button will reset the current alarm and return the Stovetop Monitor to the NORMal state, from where it will advance according to the latest temperature readings.

This App is written in MIT App Inventor 2 and has been built and tested for Android devices only. See the “Stovetop Monitor Build and Install Manual” for installation and configuration information.

See the “Overview and Theory of Operation Manual” in this repository for further information about the operation of the Stovetop Monitor.

STOVETOP MONITOR PARTICLE CONSOLE.

The information and function of the Stovetop Monitor App is also available to the user via the Particle Console. The Particle Console is web-based and can be accessed from any device that runs a web browser:

> https://console.particle.io/

The user must log in to their Particle Console with their Particle email and password. Once in the Particle Console, the user must select “My devices” and then select the Particle device (Photon 2) of their Stovetop Monitor from the list provided by Particle. The Particle Console then displays a status screen for that device. The lower right hand side of this screen contains the controls to obtain data from the device and to activate the Reset function, as depicted in figure 3, below.

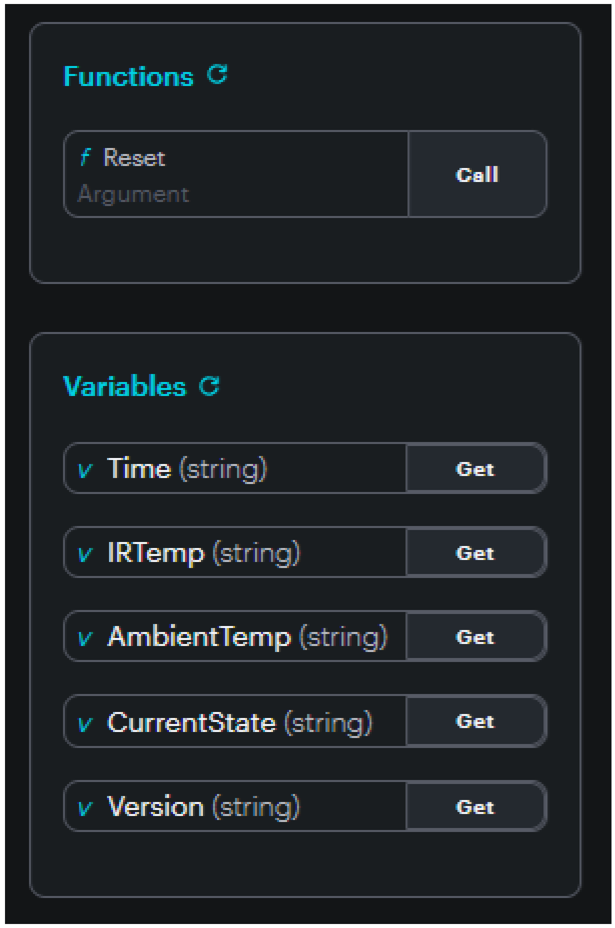

Figure 3. Particle Console Stovetop Monitor Data and Controls.

Referring to Figure 3: there is one function, called “Reset” listed under “Functions”. Clicking on “Call” is the same as tapping on “Reset Alarm” on the App. The Stovetop Monitor will reset the current alarm and return the Stovetop Monitor to the NORMal state, from where it will advance according to the latest temperature readings. No argument is needed for this function and anything typed into the “Argument” field will be ignored.

Referring to Figure 3: there are 5 “Variables” that are accessible from the Console. These are the same 5 variables as are displayed on the App screen when the “Get Data” button is tapped. However, each variable must be retrieved individually by tapping the “Get” button by each of the variable names.

See the “Overview and Theory of Operation Manual” in this repository for further information about the operation of the Stovetop Monitor.

STOVETOP CALIBRATION.

The threshold and timeout values released for this project were empirically determined by capturing various cooking sessions on a specific installation. The Stovetop Monitor firmware prints timestamped sensor data to the serial port continuously. A computer running a serial monitor program can record and save the data associated with specific cooking sessions for subsequent analysis and tuning of the threshold and timeout values for any particular installation. The file “ElectricStovetopConstants.h” can then be edited with installation specific values and saved under a different name in the same directory:

> https://github.com/TeamPracticalProjects/Stovetop_Monitor/blob/main/Software/StovetopMonitor/src/ElectricStovetopConstants.h

Line 73 of the main firmware file can then be edited to #include the new calibration data file:

> https://github.com/TeamPracticalProjects/Stovetop_Monitor/blob/main/Software/StovetopMonitor/src/StovetopMonitor.ino

Serial data from the firmware is written to the USB port of the Photon 2 microcontroller. This USB port is normally used just to power the Stovetop Monitor electronics. However, a laptop computer may be connected to this USB port and supply not only power but also receive serial data for recording and subsequent analysis. Any serial monitor program may be used for this purpose (e.g. Arduino serial monitor, VS Code serial monitor, Putty, etc.). The baud rate is 9600 baud.

Figure 4 shows a saved recording of serial data from the Stovetop Monitor using Putty. The Stovetop Monitor was reset when switching from a USB power supply to the laptop. Putty was opened after the Photon 2 reset and began running the code.

The message: “Stovetop Sensor Test … Ready for testing …” is printed during Setup(), then each new reading of data is printed on a timestamped line for as long as recording is desired. The recorded data was then stored in a file on the laptop and logged (with the test conditions under which it was obtained) for future analysis.

```
Stovetop Sensor test
Ready for testing ...
Time = 0.25 minutes     Ambient = 71.94*F       Object = 71.44*F
Time = 0.50 minutes     Ambient = 72.09*F       Object = 71.11*F
Time = 0.75 minutes     Ambient = 72.16*F       Object = 71.51*F
Time = 1.00 minutes     Ambient = 72.23*F       Object = 71.51*F
Time = 1.25 minutes     Ambient = 72.34*F       Object = 71.55*F
Time = 1.50 minutes     Ambient = 72.34*F       Object = 1899.59*F
Time = 1.75 minutes     Ambient = 72.45*F       Object = 72.05*F
Time = 2.00 minutes     Ambient = 72.48*F       Object = 72.52*F
Time = 2.25 minutes     Ambient = 72.55*F       Object = 72.95*F
Time = 2.50 minutes     Ambient = 72.55*F       Object = 1899.59*F
Time = 2.75 minutes     Ambient = 72.59*F       Object = 73.31*F
Time = 3.00 minutes     Ambient = 72.63*F       Object = 73.67*F
Time = 3.25 minutes     Ambient = 72.66*F       Object = 73.99*F
Time = 3.50 minutes     Ambient = 72.77*F       Object = 74.32*F
Time = 3.75 minutes     Ambient = 72.81*F       Object = 74.53*F
Time = 4.00 minutes     Ambient = 72.84*F       Object = 74.75*F
Time = 4.25 minutes     Ambient = 72.88*F       Object = 75.18*F
Time = 4.50 minutes     Ambient = 73.06*F       Object = 75.40*F
Time = 4.75 minutes     Ambient = 73.06*F       Object = 75.69*F
Time = 5.00 minutes     Ambient = 73.06*F       Object = 75.83*F
Time = 5.25 minutes     Ambient = 73.09*F       Object = 76.15*F
Time = 5.50 minutes     Ambient = 73.17*F       Object = 76.33*F
Time = 5.75 minutes     Ambient = 73.20*F       Object = 76.59*F
Time = 6.00 minutes     Ambient = 73.17*F       Object = 76.87*F
Time = 6.25 minutes     Ambient = 73.27*F       Object = 77.23*F
Time = 6.50 minutes     Ambient = 73.38*F       Object = 77.41*F
Time = 6.75 minutes     Ambient = 73.38*F       Object = 77.56*F
Time = 7.00 minutes     Ambient = 73.42*F       Object = 77.99*F
Time = 7.25 minutes     Ambient = 73.45*F       Object = 77.99*F
Time = 7.50 minutes     Ambient = 73.49*F       Object = 78.21*F
Time = 7.75 minutes     Ambient = 73.63*F       Object = 78.64*F
```

Figure 4. Serial Port Data from Stovetop Monitor.

Note that the recorded data is “raw” – unedited to remove the 1899.59 *F errors.

Once the data has been stored, it can be reviewed to understand the limits of the cooking session that was used to generate it:

- Peak IR temperature and peak ambient temperature captured during cooking.
- Time to peak temperature readings.
- Temperature range recorded during cooking interval.
- Time while cooking.
- Time at which heat was lowered or removed.
- Time too cool down to WARMing range (where IR temperature – ambient temperature is deemed to be too low for cooking).
- Time to cool down to NORMal range (where IR temperature is =< ambient temperature).

Note that there are a single set of thresholds and timeouts to cover all cooking scenarios. Therefore, a number of regular cooking sessions, as well as “contrived” scenarios (such as a “naked burner” left on various heat settings), must be captured and analyzed before a reasonable set of constants can be properly determined.

We recommend that thresholds and timeouts be set to favor a reasonable number of false positive alarms, on the basis that false positives are annoying but false negatives can be dangerous!